Aden Guo

Geomatics engineering graduate with hands-on experience in photogrammetry, geospatial data analysis, and field survey workflows. Builds practical data and simulation pipelines using Python/C++ and communicates results through technical reporting and presentations.

Engineering Capstone Project Lead · University of Calgary

Led a team project evaluating surveying methods for disc golf course mapping and delivered final technical findings.

- Compared RTK GNSS, range finder, and odometer techniques to assess mapping accuracy and efficiency.
- Operated drones for photogrammetric capture and executed spatial intersection workflows for 3D modeling.
- Processed and validated geospatial datasets with photogrammetry tools and Python-based analysis pipelines.
- Authored and presented the final report at the SaGES Project Presentation.

Shift Lead / Team Trainer · Gongcha

Managed high-volume shift operations while training team members on standardized service workflows.

- Trained and mentored staff to improve consistency and teamwork.
- Balanced service demands and operational tasks under time constraints.
- Strengthened team coordination, problem-solving, and day-to-day execution.

Bachelor of Science · Geomatics Engineering

Graduated

- Photogrammetry, 3D modeling, and digital imaging workflows
- Geospatial data analysis, error modeling, and quality control
- Spatial database design, SQL, and geospatial data mining
- Project management, technical reporting, and collaborative problem-solving
- Applied geomatics workflow through capstone and ENGO 500 field survey camp

Programming & Data Science

Python, JavaScript, TypeScript, SQL, MATLAB, C++, NumPy, Pandas, SciPy, scikit-learn, PyTorch,

OpenCV

GIS & Survey Tools

ArcGIS, QGIS, Google Earth Engine, PostGIS, GeoPandas, GDAL/PROJ, RTK GNSS workflows, Total Stations, Digital leveling

Workflow & Version Control

Git, GitHub, VS Code, Jupyter, Anaconda, virtualenv, Docker (basic)

Sensor Simulation & Estimation

IMU/GNSS simulation, Extended Kalman Filter (EKF)

Proprietary & Survey Software

AutoCAD, Civil 3D, MicroStation, Agisoft Metashape, Trimble Business Center (TBC)

Photogrammetry Pipeline - Bundle Adjustment & Spatial Intersection

Modular photogrammetry workflow using synthetic and real-world datasets for spatial intersection and bundle adjustment.

- Implemented camera models, pose estimation, and 3D reconstruction with C++ and Python.
- Built residual, optimization, and quality-assessment components to improve reconstruction accuracy.

ML-Driven Sensor Simulation + EKF Sensor Fusion

Simulation framework for IMU/GNSS trajectory generation with machine learning and EKF fusion.

- Developing RNN/Transformer-based models for trajectory simulation.
- Integrating EKF for real-time estimation of pose, velocity, and orientation.
- Designing modular training, visualization, and evaluation pipelines.